

INDUSTRIAL INTERNSHIP REPORT

on

GRAPHICS DESIGN AND PRODUCT PROTOTYPING FOR A CONSUMER TOY BRAND

A study of packaging, dieline design, prototyping and design-for-manufacturing practice at InventoJoy,
Intelligentia Labs Pvt. Ltd.

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Roll No.: 2410030504

Section: 3CSE26

Under the guidance of

Internal Guide: [Name and designation of faculty guide]

External Guide: [Name and designation of company mentor]

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

[NAME OF UNIVERSITY / INSTITUTION]

[Campus and location]

Academic Session [2025–26]

CERTIFICATE OF INTERNSHIP COMPLETION

ATTACH HERE: The internship completion certificate issued by Intelligentia Labs Pvt. Ltd. on company letterhead. Most universities require this as an original or attested copy. If the certificate is a separate physical document, insert a scanned image on this page or state that it is attached as an annexure.

This is to certify that Mr. Nitin Kumar Choudhary, Roll No. 2410030504, Section 3CSE26, has satisfactorily completed an industrial internship in the capacity of Graphics and Prototyping Intern with Intelligentia Labs Pvt. Ltd. (InventoJoy), Greater Noida, from 26 March 2026 to [end date].

During the period of the internship, the work and conduct of the intern were found to be [satisfactory / as recorded by the organisation].

Signature and Seal

[Name], [Designation]

Intelligentia Labs Pvt. Ltd.

Date: _____

CERTIFICATE OF APPROVAL

This is to certify that the internship report entitled "Graphics Design and Product Prototyping for a Consumer Toy Brand", submitted by Nitin Kumar Choudhary, Roll No. 2410030504, Section 3CSE26, is a record of bona fide work carried out by him under our supervision, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

The report has not been submitted, in whole or in part, for the award of any other degree or diploma.

Internal Guide

[Name and designation]

Head of Department

[Name and designation]

External Examiner

DECLARATION

I, Nitin Kumar Choudhary, bearing Roll No. 2410030504 of Section 3CSE26, hereby declare that the internship report entitled "Graphics Design and Product Prototyping for a Consumer Toy Brand" is an authentic record of the work carried out by me during my industrial internship with Intelligentia Labs Pvt. Ltd. (InventoJoy), Greater Noida, commencing 26 March 2026, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

I further declare that the observations, methods and conclusions recorded in this report are my own, that all sources consulted have been acknowledged, and that no portion of this work has been submitted for the award of any other degree or diploma at this or any other institution.

I confirm that the contents of this report have been prepared in accordance with the confidentiality undertaking given to Intelligentia Labs Pvt. Ltd. No proprietary artwork, source file, commercial information or unreleased product detail has been reproduced without the express permission of the organisation.

Nitin Kumar Choudhary

Roll No. 2410030504 | Section 3CSE26

Place: _____ Date: _____

ACKNOWLEDGEMENT

I wish to record my sincere gratitude to Intelligientia Labs Pvt. Ltd. for the opportunity to undertake an industrial internship with the InventoJoy brand, and for the access and trust extended to me during the engagement.

I am grateful to [name and designation], Director, InventoJoy, for approving the appointment and for the direction provided during the internship.

I am particularly indebted to my reporting manager, [name and designation], whose review and guidance shaped every deliverable described in this report. The habit of physically verifying a design before defending it, which recurs throughout this document, was learned directly from that supervision.

I thank the members of the product and manufacturing teams at InventoJoy for their patience in explaining constraints that were not obvious to me, and for the cross-functional reviews that improved the work substantially.

I extend my thanks to [name], Head of the Department of Computer Science and Engineering, and to my internal guide [name], for their academic supervision and for their guidance in the preparation of this report.

Finally, I thank my family and my peers for their support throughout the period of this internship.

Nitin Kumar Choudhary

Roll No. 2410030504 | Section 3CSE26

ABSTRACT

This report documents an industrial internship undertaken as Graphics and Prototyping Intern at Intelligentia Labs Pvt. Ltd., the company operating the consumer toy brand InventoJoy, at its office in EPIP Site-5, Greater Noida. The engagement commenced on 26 March 2026 as a full-time work-from-office position with a minimum contracted duration of three months.

The problem addressed is characteristic of a growing consumer manufacturing brand. Design work in such an environment cannot be judged on visual merit alone: a packaging design is simultaneously a structural artefact, a logistics variable, a regulatory instrument and a retail communication device. A graphic approved on screen may fail when the board is folded; a carton that presents well on a shelf may be uneconomic to ship; an aesthetically resolved layout may omit a statutory warning. The internship was concerned with satisfying these constraints together, and with doing so before production tooling is committed.

The work spans five streams derived from the appointment letter: packaging and dieline design covering boxes, inserts and stickers optimised for manufacturing, shipping and shelf display; physical prototyping and support for product testing; the design of intuitive, child-friendly assembly guides; collaboration with product and manufacturing teams on design-for-manufacturing readiness; and support for marketing creatives across digital and print. Chapter 5 documents one project end to end — the complete printed system for a STEM construction kit, comprising carton, dieline, insert tray, sticker sheet and pictorial assembly guide — so that method can be traced through a single artefact rather than inferred from a list of outputs.

The methodology is an iterative, gated nine-stage workflow. Its defining features are that the structural template is fixed and physically folded before graphic development begins, that mandatory regulatory content is placed on the dieline before creative layout, and that verification fidelity is escalated only as far as the current question requires — a paper fold before a plotter-cut mock-up, a mock-up before a substrate comp, a comp before a vendor sample. Decisions pass through review gates with explicit exit criteria, including a design-for-manufacturing gate held jointly with the product and manufacturing teams and a pre-press checklist executed before any vendor release.

The report is explicit about the status of its claims. This was a professional design engagement rather than a research study: no controlled experiment, statistically valid user test or instrumented measurement was conducted, and Chapter 7 therefore presents qualitative observations together with a proposed evaluation framework in which every metric is labelled as not measured. Confidentiality obligations further restrict reproduction of proprietary artwork and unreleased product detail.

The principal conclusion is that in a manufacturing context the quality of a design outcome is governed less by the merit of individual creative decisions than by the order in which constraints are admitted into the process. Constraints consulted early become design parameters; the same constraints discovered late become revisions, and after tooling commitment they become production losses.

Keywords

Industrial internship; graphic design; packaging design; dieline; structural packaging; prototyping; additive manufacturing; design for manufacturing; instructional design; assembly guide; pre-press; print production; toy safety and labelling; shelf display; logistics optimisation; version control; consumer product design.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
3D	Three-Dimensional
BIS	Bureau of Indian Standards

Abbreviation	Expansion
CAD	Computer-Aided Design
CMYK	Cyan, Magenta, Yellow and Key (black) — subtractive print colour model
DAM	Digital Asset Management
DFM	Design for Manufacturing
FBB	Folding Box Board
FDM	Fused Deposition Modelling
GSM	Grams per Square Metre
NCC	Non-Compete Clause
NDA	Non-Disclosure Agreement
PPO	Pre-Placement Offer
RGB	Red, Green and Blue — additive screen colour model
RTE	Reverse Tuck End (carton construction)
SKU	Stock Keeping Unit
SLA	Stereolithography
STE	Straight Tuck End (carton construction)
UPSIDA	Uttar Pradesh State Industrial Development Authority
WS	Work Stream (as used in this report)

NOTE ON COMPLETION

This document contains marked placeholders wherever a fact specific to the student's own engagement is required. Grey boxes headed "TO COMPLETE" or "ACTION REQUIRED" must be replaced with the student's own account or deleted; dashed boxes marked "[INSERT IMAGE HERE]" mark the figure positions listed in Appendix D. No placeholder should remain in the submitted version. The report deliberately states no measured result that was not recorded during the internship; statements marked as proposed should not be converted into claimed results.

CHAPTER 1: INTRODUCTION

1.1 Introduction

An industrial internship is the point at which academic preparation meets the constraints of commercial production. In a university setting a design problem is bounded and evaluated against a rubric. In an operating company the same problem is bounded by cost per unit, tooling limits, material availability, shipping dimensions, retail shelf conventions, statutory labelling and a delivery date already promised to a customer. The transition between these two environments is the educational value of an internship, and it is the subject of this report.

This report documents an industrial internship undertaken as Graphics and Prototyping Intern at Intelligentia Labs Pvt. Ltd., which owns and operates the toy brand InventoJoy. The engagement was based at the company office at E-003, First Floor, EPIP Site-5, UPSIDA, Greater Noida – 201306, as a full-time work-from-office position commencing 26 March 2026 with a minimum contracted duration of three months.

The role sits between two disciplines that are usually taught separately but are inseparable in toy manufacturing: graphic design, which governs how a product communicates with a child and with a purchasing parent, and physical prototyping, which governs whether the designed object can actually be produced, packed, shipped, opened and assembled. A toy that is beautifully illustrated but arrives in a crushed carton has failed; a carton that survives transit but cannot be opened by an eight-year-old has also failed. The internship was concerned with the space between those two failures.

1.2 Objectives of the Internship

The objectives below derive from the responsibilities defined in the appointment letter and form the framework against which the work in Chapters 5 and 6 was planned and reviewed.

1. To apply graphic design fundamentals to toy packaging and printed collateral — boxes, dielines, inserts and stickers.
2. To develop working knowledge of structural packaging, specifically dielines optimised for manufacturing, shipping and shelf display.
3. To assist in building physical prototypes and in testing products against functional and durability expectations.
4. To design instruction manuals and assembly guides that are intuitive, largely language-independent and appropriate to the comprehension level of a child user.
5. To collaborate with the product and manufacturing teams so that packaging and products submitted for production are Design-for-Manufacturing ready.
6. To support the marketing function with creatives for digital and print deployment.
7. To develop professional competencies: version control of design files, structured vendor handover, responding to review feedback and working to production deadlines.

1.3 Scope of the Report

The report covers the technical background necessary to understand the work, the organisational context, the methodology followed, a documented project, the wider work undertaken, and the competencies developed. Chapter 5 presents one project in full detail as a representative case, because a single project traced end to end demonstrates method more convincingly than a list of outputs.

The report does not cover commercial performance, supplier identities or unreleased product concepts. Where a work item cannot be described specifically for confidentiality reasons, the class of problem and the method applied are described instead.

1.4 Limitations of the Report

- Confidentiality. The appointment letter imposes a Non-Disclosure Agreement covering source files, assets and operational processes — precisely the outputs of this role. Proprietary artwork is therefore described, redacted or generalised rather than reproduced.
- Measurement. This was a design engagement, not a research study. No controlled experiment, statistically valid user test or instrumented measurement was conducted. Chapter 7 presents qualitative observations and a proposed evaluation framework, not measured results.
- Attribution. Design work in a studio is collaborative and supervised. Deliverables described here were executed under the direction of the reporting manager and reviewed by the product and manufacturing teams.

1.5 Organisation of the Report

Table 1: Organisation of the report

Chapter	Content
Chapter 2	The host organisation, the brand, the working environment, terms of engagement and the confidentiality position of this report.
Chapter 3	Technical background: graphic and packaging fundamentals, dielines, print production, design for manufacturing, prototyping methods, instructional design and toy labelling.
Chapter 4	The design and prototyping methodology followed, from brief to vendor handover, with its review gates.
Chapter 5	A representative project documented in full: packaging, insert, sticker sheet and assembly guide for a STEM construction kit.
Chapter 6	The wider work undertaken across the five work streams defined by the appointment letter.
Chapter 7	Observations, challenges, proposed evaluation framework and learning outcomes.
Chapter 8	Conclusion and future scope.

CHAPTER 2: ORGANISATION PROFILE

2.1 Intelligentia Labs Pvt. Ltd. and InventoJoy

Intelligentia Labs Pvt. Ltd. is the corporate entity under which the internship was undertaken, operating from EPIP Site-5, an industrial area administered by the Uttar Pradesh State Industrial Development Authority in Greater Noida. The location places the design function close to the manufacturing and vendor ecosystem of the National Capital Region, which matters for a role requiring frequent physical review of samples, board stocks and printed proofs.

InventoJoy is the company's toy brand. The appointment letter describes it as a growing brand, and the responsibilities attached to the internship — packaging, dielines, inserts, stickers, prototypes, assembly guides and marketing creatives — are characteristic of a brand at the stage where product range and brand system are being built concurrently rather than sequentially.

This growth stage changes how a designer works. In a mature brand most design activity is application of an established system to a new product. In a growing brand it is partly application and partly system-building: templates, dieline libraries, colour standards and instructional iconography are created as a by-product of individual projects. Recognising this early was a useful adjustment, because a file built as a reusable template repays itself across the range while a file built as a one-off does not.

TO COMPLETE: Add a short factual paragraph on the company — year of incorporation, team size, product categories, target age bands and sales channels. Take these from the company website or ask your reporting manager; do not estimate.

2.2 Working Environment and Reporting

The engagement was a work-from-office position with working hours from 10:00 to 19:30, extendable according to operational requirements. Co-location matters for this role because prototyping is physical: board samples, printed proofs, folded mock-ups and assembled toys must be handled, folded, dropped and shown to colleagues. A dieline review happens around a table with the object present, not over a screen share.

Work was assigned and reviewed by the reporting manager, with periodic performance review as provided in the appointment letter. Three informal review mechanisms operated continuously and accounted for most feedback received: desk review of work in progress, physical review conducted around a mock-up or prototype, and cross-functional review by the product or manufacturing team.

2.3 Terms of Engagement

Table 2: Summary of internship engagement terms

Parameter	Detail
Organisation and brand	Intelligentia Labs Pvt. Ltd. — InventoJoy
Designation	Graphics and Prototyping Intern
Commencement	26 March 2026
Minimum duration	3 months, with a 7-day probation period
Working hours	10:00 to 19:30, subject to operational requirements
Mode and location	Work from office — E-003, First Floor, EPIP Site-5, UPSIDA, Greater Noida – 201306
Review	Periodic performance review by the reporting manager
Notice period	30 days

Parameter	Detail
Confidentiality	Non-Disclosure Agreement covering customer data, business strategies, source files, assets and operational processes
Restrictive covenant	Non-Compete Clause — 6 months post-departure, e-commerce and toy sector, within India
Progression	Pre-Placement Offer may be considered subject to performance

2.4 Confidentiality Position of This Report

The Non-Disclosure Agreement expressly covers source files, assets and operational processes. Since these are the outputs of a graphics and prototyping role, the obligation determines what an academic report may legitimately contain. The position adopted is that method, reasoning, workflow, checklist and standard are described in full, because these are transferable professional knowledge; specific artwork, production dielines, vendor identities, costing and unreleased product detail are omitted, generalised, or reproduced only with the organisation's consent.

ACTION REQUIRED: Before submission, obtain written permission from your reporting manager for every image you insert. Ask specifically whether already-released product packaging may be shown. Where permission is refused, substitute a generic hand-drawn diagram illustrating the same principle and note this under the caption.

CHAPTER 3: TECHNICAL BACKGROUND

3.1 Graphic Design Fundamentals in a Packaging Context

Visual hierarchy is the ordering of elements so a viewer perceives them in an intended sequence, and on a toy carton it is commercially decisive because a shopper gives a single facing very little attention. The conventional hierarchy places the product name and its illustration first, the brand mark second, the age grading and key claim third, and secondary information last, built from size, weight, contrast, isolation and position acting together.

Colour signals the age band and differentiates products within a range, and must survive translation from screen to print. Typography carries a legibility obligation beyond aesthetics: statutory text is subject to minimum size requirements, and small type reversed out of busy illustration can be compliant in point size while being unreadable in practice. The working rule applied throughout was to set all mandatory text on a plain, high-contrast ground.

3.2 Packaging Functions

Table 3: Functions of toy packaging and their design implications

Function	Design implication
Containment	Internal volume, insert or tray design, immobilisation of parts to prevent rattle and abrasion in transit
Protection	Board grade and thickness, flute selection, cushioning, drop performance
Communication	Front-panel hierarchy, product imagery, feature callouts, brand system
Compliance	Age grading, warning statements, manufacturer details, country of origin, conformity marking, barcode
Retail performance	Shelf facing dimensions, stackability, hang-tab or shelf-ready format, planogram footprint
Logistics efficiency	Master carton fit, cube utilisation, pallet pattern, dimensional weight for e-commerce

3.3 Dielines and Structural Packaging

A dieline is a flat, dimensioned vector template defining how a sheet of board is cut, creased and assembled into a three-dimensional package. It is the interface between designer and die-maker, and the most error-prone artefact in packaging production, because an error in it surfaces only after tooling has been cut. Each line type is conventionally assigned a distinct spot colour so it can be separated and suppressed at output; misassigning a line type is a frequent cause of production failure.

Table 4: Dieline line types and carton constructions

Element	Purpose or characteristic	Convention or application
Cut line	Where the board is cut through	Solid, dedicated spot colour
Crease line	Where the board is scored to fold	Dashed, separate spot colour
Perforation	Tear-open features and opening strips	Dash-dot, separate spot colour
Bleed	Artwork extended beyond the cut line	Typically 3 mm on every cut edge
Safety zone	Inner margin for critical content	Typically 3–5 mm inside cut and crease lines
Reverse tuck end	Tucks fold from opposite directions; economical die	Lightweight boxed items and small kits

Element	Purpose or characteristic	Convention or application
Straight tuck end	Both tucks fold from the same direction; cleaner front face	Where front-panel appearance is critical
Auto-bottom	Pre-glued base erecting automatically; faster packing, stronger base	Heavier contents, higher-volume lines
Window box	Die-cut aperture, usually with a clear film patch	Where product visibility drives purchase
Corrugated mailer	Self-locking single-wall construction for transit	Direct-to-consumer e-commerce shipping

[INSERT IMAGE HERE] — Export a generic dieline (not a proprietary one) and annotate the line types in Illustrator or PowerPoint.

Figure 1: Annotated dieline showing cut, crease, bleed and safety zones

3.4 Print Production

Artwork is authored on screen in an additive RGB space but reproduced on press in a subtractive CMYK space, supplemented by spot colours. The conversion is not lossless: saturated blues, greens and oranges commonly fall outside CMYK gamut and shift. The practice adopted was to work in the destination colour space from the outset for brand-critical elements and to specify brand colours as spot references with documented CMYK equivalents. Raster artwork for offset printing is prepared at approximately 300 pixels per inch at final size, with effective resolution checked at placed size rather than at nominal file dimensions.

Substrate and finish selection is a cost and performance decision. Folding boxboard grades differ in fibre composition, reverse-face whiteness and stiffness; corrugated flute profiles trade print quality against crush resistance. Lamination, coating, foiling and embossing each add cost and lead time and interact with glue areas, which must be kept free of varnish and lamination.

3.5 Design for Manufacturing

Design for Manufacturing is the practice of designing so a product can be produced reliably and economically with the intended process, tooling and materials. In this role it applied in two directions at once. For the toy, it meant draft angles and wall thickness for moulded parts, part consolidation to reduce assembly labour, tolerance stack-up across mating parts and avoidance of undercuts. For packaging, it meant whether a dieline nests efficiently on the press sheet, whether glue flaps suit the board grade, whether the carton can be erected on the available packing line, and whether the finished pack fits the master carton without void space.

The organisational significance is that DFM converts a design review from a subjective discussion into a constraint check — one to be run before submitting work rather than after.

3.6 Prototyping Methods

A prototyping method is selected according to the question being asked. A prototype built to answer a proportion question need not function; one built to answer a durability question need not be attractive. Confusing the two wastes time and produces false confidence.

Table 5: Prototyping methods and the questions they answer

Method	Question it answers	Typical turnaround
Printed and hand-folded paper mock-up	Proportion, panel layout, artwork across folds, legibility at true size	Under an hour

Method	Question it answers	Typical turnaround
Plotter-cut board mock-up	Structural behaviour, closure fit, board stiffness, stacking	Hours
FDM 3D print	Form, fit, ergonomics, part interference	Hours to a day
Vendor pre-production sample	Whether the production process can deliver this consistently	Weeks

The escalation principle followed throughout was to answer as many questions as possible at the cheapest fidelity before escalating. A vendor sample is authoritative but slow and expensive; a folded paper mock-up costs a sheet of paper and fifteen minutes, and in practice caught most layout and proportion errors.

3.7 Instructional Design for Assembly Guides

An assembly guide for a children’s product is a technical communication problem with a demanding audience: a child with limited reading ability, a parent under time pressure, or a user who does not read the printed language. The design response, established in the toy and flat-pack furniture industries, is to minimise dependence on text.

- Language-independent pictograms carry the instruction; text is supplementary rather than load-bearing.
- One action per step. Combining two actions into a single illustrated step is the commonest cause of assembly error.
- A consistent viewpoint across steps, since changing the viewing angle forces mental re-orientation.
- A parts inventory at the outset, each part illustrated at consistent scale with an identifier used throughout.
- Safety and adult-supervision notices placed before the step they relate to, not only on a cover.

3.8 Toy Safety and Labelling

Toy safety in India is governed by a quality-control regime requiring conformity to prescribed Indian Standards and the corresponding conformity marking, covering mechanical and physical properties, flammability and migration of certain elements. Products for export are additionally assessed against destination-market standards. For a graphics role the regime manifests mainly through labelling: age grading, warning statements including small-parts warnings where applicable, manufacturer or importer identification, country of origin and the conformity mark. These are not discretionary content; correct practice, followed throughout, is to place them on the dieline before creative layout begins rather than fitting them into whatever space remains.

VERIFY BEFORE SUBMISSION: Standards and quality-control orders are revised periodically. Confirm the exact standard designations, editions and marking requirements applicable to InventoJoy’s products with your reporting manager or the company’s compliance contact and cite them here. Do not state standard numbers you have not verified — an incorrect citation is worse than a general statement.

CHAPTER 4: METHODOLOGY

4.1 Overall Workflow

The workflow followed nine stages. Its two defining features are that the dieline is fixed and physically folded before artwork begins, and that verification fidelity is escalated only as far as the current question requires.

1. Brief intake: objective, product, audience, channel, constraints, deadline and approval path.
2. Research and reference: category conventions, existing brand assets, competitor shelf presence, applicable compliance requirements.
3. Concept development: divergent options generated at low fidelity and evaluated against the brief rather than by preference.
4. Structural definition: dieline selected or constructed, with bleed, safety zones, glue areas and closure mechanism resolved.
5. Artwork development: graphics applied within the fixed template, mandatory content placed first.
6. Physical mock-up and iteration: print, cut, fold, assess; iterate on proportion, legibility and structural behaviour.
7. DFM and cross-functional review: constraint check with the product and manufacturing teams.
8. Revision and pre-press: feedback incorporated, pre-press checklist executed.
9. Handover and archive: vendor handover pack prepared, approval recorded, version archived.

[INSERT IMAGE HERE] — Draw the nine stages as a flowchart with two feedback loops marked: mock-up iteration returning to artwork development, and cross-functional review returning to structural definition.

Figure 2: End-to-end design and prototyping workflow

4.2 Brief Intake

An incompletely specified brief is the most expensive failure mode in a design cycle because it is discovered only after production effort has been spent. The intake practice was to confirm a defined parameter set in writing before starting: objective and success criterion; product, variant and whether dimensions are final; audience age band and purchase decision-maker; channel; whether the dieline is fixed, adapted or new; the full content inventory including copy, claims, statutory text, barcode and certification marks; production constraints of substrate, colours, finishes, vendor and quantity; and the deadline with its approval path.

4.3 Review Gates

Table 6: Review gates and exit criteria

Gate	Reviewer	Exit criterion
Brief confirmation	Reporting manager	All intake parameters resolved in writing
Concept selection	Reporting manager	One direction selected against brief criteria
Structural sign-off	Reporting manager; manufacturing	Dieline physically folded and dimensions verified against the product
Design review	Reporting manager	Hierarchy, brand consistency and mandatory content approved
DFM review	Product and manufacturing	Every constraint satisfied or formally accepted

Gate	Reviewer	Exit criterion
	teams	as a deviation
Pre-press check	Designer; vendor pre-flight	All checklist items cleared (Appendix C)
Final approval	Reporting manager	Approved version identifier recorded in writing

4.4 Version Control and Handover

All design files followed the pattern `PRODUCTCODE_ARTEFACT_VARIANT_vNN_STATUS.ext`, where the status token distinguishes WIP, REVIEW, APPROVED and PRODUCTION states. Only files carrying the PRODUCTION token were released to a vendor, which prevents an unapproved file reaching production — a version-control failure being a commoner cause of production error than a design failure. Version numbers increment on every substantive change and are never overwritten; superseded versions are archived rather than deleted.

Handover to a vendor comprised the production file with fonts outlined and links packaged, the dieline on a separate suppressible layer, a flattened reference PDF showing intended appearance, a specification sheet stating substrate, finishes, colours and quantity, and a written statement of the approved version identifier.

CHAPTER 5: PROJECT — PACKAGING AND ASSEMBLY GUIDE SYSTEM FOR A STEM CONSTRUCTION KIT

READING NOTE: This chapter documents one project end to end as a representative case of the work undertaken, so that method can be traced through a single artefact rather than inferred from a list of outputs. The product is referred to by an anonymised code and the specification values stated are the design values used in this case. Before submission, align the product name, dimensions, board grades and outcomes with your own project, and replace the figure placeholders with your own images.

5.1 Project Brief

The project covered the complete printed system for a STEM construction kit, referred to here as product code IJ-SC-01: a 42-part snap-together building set for the six-plus age band, sold both through retail shelf and through e-commerce marketplaces. The deliverables were a primary carton with dieline, an internal insert tray, a sticker sheet, a pictorial assembly guide, the master shipping carton specification, and a set of listing creatives.

Two constraints shaped every subsequent decision. First, the product sells through two channels with opposing requirements: retail shelf rewards a large, visually assertive facing, while e-commerce penalises volume through dimensional weight. Second, a 42-part kit with small components triggers small-parts labelling obligations and requires an insert that immobilises parts so they neither rattle nor abrade in transit.

Table 7: Project requirement summary

Parameter	Requirement
Product	IJ-SC-01 — snap-together STEM construction kit, 42 parts
Age band	6 years and above
Channels	Retail shelf and e-commerce marketplace
Primary pack	Printed folding carton with product visibility
Internal	Insert or tray immobilising all parts; sticker sheet
Documentation	Pictorial assembly guide, minimal text dependence
Compliance	Age grading, small-parts warning, manufacturer details, country of origin, conformity mark, barcode
Transit	Master carton fit; survive e-commerce single-parcel handling
Constraint	Minimise pack volume without weakening shelf facing

5.2 Structural Design and Carton Selection

Structural definition preceded artwork, per Section 4.1. Three constructions were evaluated against the requirement set.

Table 8: Carton construction options evaluated

Option	Advantages	Disadvantages	Assessment
Reverse tuck end	Lowest die and material cost; simple to erect by hand	Weaker base under load; opposing tuck directions look untidy on the front face	Rejected — base strength inadequate for a parts-heavy kit
Auto-bottom with window	Strong pre-glued base; fast to erect; window gives product visibility on shelf	Higher die cost; window patch adds a process step and complicates recycling	Selected

Option	Advantages	Disadvantages	Assessment
Two-piece rigid lid and base	Premium presentation and reusable as storage	Substantially higher unit cost and greater pack volume	Rejected — cost and dimensional weight

The auto-bottom construction with a die-cut window was selected. The pre-glued base carries the load of a parts-heavy kit without the base failure characteristic of a tuck end, and it erects on the packing line without manual base folding. The window was sized to reveal the assembled hero model rather than the loose parts, which addresses the shelf requirement without enlarging the pack.

Internal dimensions were set from the insert tray envelope plus a tolerance allowance, and board thickness was accounted for at every fold rather than only at the outer dimensions — an omission that produces a carton which measures correctly flat but binds when erected. Bleed and safety zones were applied to the vendor's specification, and cut, crease and glue areas were assigned to their conventional layers and spot colours.

[INSERT IMAGE HERE] — *Insert your flat dieline showing cut, crease, glue and window aperture on separate layers.*

Figure 3: IJ-SC-01 primary carton — flat dieline with line types annotated

5.3 Insert Tray Design

A 42-part kit presents a containment problem that the outer carton cannot solve. Loose parts migrate in transit, abrade printed surfaces, and produce a rattle that reads at the point of sale as breakage. Three approaches were considered: a vacuum-formed plastic tray, a moulded pulp tray, and a die-cut folding board insert.

The folding board insert was selected. It is produced on the same substrate and by the same vendor as the carton, which removes a supplier and a lead time from the critical path; it is compatible with the brand's stated direction on plastic reduction; and it can be revised at the cost of a die change rather than a mould. Its disadvantage is lower part-location precision than a formed tray, which was mitigated by grouping parts into compartments by size rather than locating each part individually.

The insert was designed as a single blank folding into a compartmented tray, with a lift tab so a child can remove the tray without inverting the carton. The sticker sheet sits flat beneath the tray, where it is protected from creasing and is not the first thing encountered on opening.

[INSERT IMAGE HERE] — *Photograph the folded insert with parts loaded, alongside its flat dieline.*

Figure 4: Insert tray — dieline and folded assembly

5.4 Graphic Design and Panel Layout

Artwork was developed panel by panel on the fixed dieline rather than across the flat sheet, because a composition that balances flat can be incoherent once folded. The front-panel hierarchy was resolved first, since it carries the commercial burden: the assembled hero model and product name lead, the brand mark follows, the age grading and part count sit third, and supporting content is subordinate to all three.

The window aperture was treated as an element of the composition rather than as a hole cut into a finished layout. Positioning it low and to one side allowed the illustration and typography to occupy the optically dominant upper region while the physical product occupied the lower, which produces a stronger facing than a centred window surrounded by a frame of artwork.

Side panels carried feature callouts and the build examples that extend perceived value; the rear panel carried the parts inventory, statutory content and barcode. Two disciplines were applied without exception: no critical element within the safety zone of any cut or crease line, and mandatory content treated as fixed with creative content arranged around it.

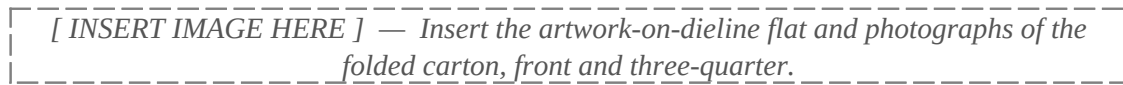


Figure 5: IJ-SC-01 — applied artwork on the dieline and assembled carton views

5.5 Mandatory Content Placement

Statutory and regulatory content was placed on the dieline before creative layout began. For a small-parts construction kit in this age band the content set comprised the age grading, the small-parts warning, manufacturer and importer identification, country of origin, the conformity mark, the barcode with its quiet zones, and the parts inventory.

Placing this content first has a consequence worth stating plainly: it constrains the creative layout, and it is meant to. When mandatory content is placed last it occupies residual space, is set at minimum size on a busy ground, and becomes the weakest element on the pack. The rear panel was therefore laid out on a plain ground with the statutory block given a defined, uninterrupted region.

VERIFY: Confirm the exact wording, minimum type size and mark proportions required for your product with the company's compliance contact before finalising this section.

5.6 Prototyping and Iteration

Verification followed the escalation principle in Section 3.6. Each iteration was built to answer a specific question, and fidelity rose only when the question required it.

Table 9: Prototype iteration record for IJ-SC-01

Iteration	Method	Question addressed	Outcome
v01	Desktop print, hand cut and folded	Are proportions correct? Does artwork survive the folds?	Front-panel typography fell within the crease safety zone; hierarchy competing between name and hero model
v02	Desktop print, hand cut and folded	Do the revised layout and window position work at true size?	Window position improved facing; insert tray envelope found to exceed the internal dimension
v03	Plotter-cut on representative board	Does the construction hold? Does the insert seat correctly?	Internal dimension corrected for board thickness; lift tab added to the tray
v04	Plotter-cut with parts loaded	Do parts remain immobilised? Does the pack rattle when shaken?	Compartment depths adjusted; two compartments merged to reduce insert complexity
v05	Comp on intended substrate and finish	Does colour reproduce as intended? Does the finish suit the positioning?	Brand colour shift observed on conversion; spot reference specified with documented CMYK equivalent
v06	Vendor pre-production sample	Can the process deliver this consistently?	To be recorded — see completion note below

TO COMPLETE: Replace the outcomes above with what your own iterations actually revealed, and record the outcome of the vendor sample. Iteration records are the most persuasive evidence in a design report because they show reasoning rather than results — be specific about what went wrong at each stage.

[INSERT IMAGE HERE] — Photograph successive mock-ups together, oldest to newest, in a single frame so iteration is visible at a glance.

Figure 6: Prototype iteration sequence, v01 to final

5.7 Assembly Guide Design

The guide was designed to the principles in Section 3.7. A 42-part kit with multiple build configurations presents a sequencing problem before it presents an illustration problem: the guide must establish the parts vocabulary, then teach the connection mechanism, then build the hero model, then present alternative builds.

The structure adopted was a parts inventory spread with every part illustrated at consistent scale and given a persistent identifier; a connection-principle page demonstrating how parts join, taught once rather than repeated; a step sequence for the hero model at one action per step and a fixed isometric viewpoint; and an alternative-builds section at lower instructional density. Safety and supervision notices were positioned ahead of the steps they relate to rather than only on the cover.

Validation was carried out by handing the guide and the parts to a colleague unfamiliar with the product and observing silently while they assembled it. This method matters because a guide is written by someone who already knows how the product goes together, and the author is therefore almost incapable of detecting an ambiguous step. Every point at which the assembler hesitated was treated as a defect in the guide, not in the assembler.

TO COMPLETE: Record what your own validation revealed — which steps caused hesitation, and what you changed. This is the single most valuable paragraph in the chapter if you completed the exercise; report it specifically.

[INSERT IMAGE HERE] — Insert the parts inventory page and a spread of consecutive assembly steps showing consistent viewpoint and one action per step.

Figure 7: Assembly guide — parts inventory and step sequence

5.8 Design for Manufacturing Review

The design passed through a joint review with the product and manufacturing teams before pre-press. The review was treated as a gate rather than a consultation: no item proceeded until each constraint was either satisfied or formally accepted as a deviation.

Table 10: DFM constraints raised and design responses

Constraint raised	Origin	Design response
Glue flap dimension marginal for the specified board grade	Manufacturing	Flap width increased; dieline re-nested to absorb the change without additional sheet area
Window patch adds a process step and a lead time	Manufacturing	Retained — accepted as a deviation on shelf-visibility grounds, with the aperture reduced to limit patch material
Insert compartment count increased assembly labour at packing	Product	Two compartments merged; parts grouped by size rather than located individually
Heavy ink coverage encroaching	Manufacturing	Artwork pulled back; glue areas kept free of

Constraint raised	Origin	Design response
on glue areas		varnish and lamination
Pack depth driving dimensional weight on e-commerce	Product	Depth reduced by revising insert compartment depths after iteration v04
Master carton fit leaving avoidable void	Manufacturing	Primary pack footprint adjusted so units nest without void space

The most consequential learning here concerned sequencing. A constraint discovered at review is a revision; the same constraint anticipated during design is simply a parameter. Over the project the practice shifted from designing and then submitting for a constraint check, to consulting manufacturing at the structural definition stage.

5.9 Pre-Press and Handover

Pre-press was executed as the fixed checklist in Appendix C rather than as review by inspection, because the errors it catches are precisely those inspection misses. Verification covered separations and spot colour count, effective resolution at placed size, bleed and safety compliance, dieline layer separation and output suppression, overprint settings, font handling, barcode magnification and quiet zones, and every item of statutory content against the approved content list. The handover pack and approval record followed Section 4.4.

5.10 Project Outcome

TO COMPLETE: State the outcome plainly — what was approved, what went to production, what was superseded and what remained open at the end of the internship. Do not claim commercial results such as sales impact, which you are not in a position to verify. Work that was shelved is not a failure and should be reported; in a commercial studio a substantial proportion of output never reaches production.

Three observations from the project generalise beyond it. The cost of a change rises sharply with the stage at which it is made, which is the entire argument for cheap early iteration. Mandatory content placed last is always placed badly. And a dieline that has never been physically folded is an untested assumption, however carefully it was constructed on screen.

CHAPTER 6: WIDER WORK UNDERTAKEN

6.1 Work Streams

The appointment letter defines five responsibility areas. Chapter 5 documents a project spanning all five; this chapter records the wider work undertaken in each.

Table 11: Mapping of appointment-letter responsibilities to work streams

Responsibility as stated	Work stream	Principal competencies
Designing toys, packaging and graphics (boxes, dielines, inserts, stickers) optimised for manufacturing, shipping and shelf display	WS-1	Vector production, structural packaging, pre-press, colour management, shelf and logistics reasoning
Creating prototypes and helping test products	WS-2	Model making, 3D printing, materials handling, structured observation
Designing intuitive, child-friendly instruction manuals and assembly guides	WS-3	Technical illustration, pictogram systems, information sequencing
Collaborating with product and manufacturing teams for DFM-ready solutions	WS-4	Constraint analysis, technical communication, review discipline
Supporting marketing creatives for digital and print platforms	WS-5	Small-format composition, adaptation systems, asset management

6.2 Record of Assignments

Table 12: Record of assignments undertaken (to be completed)

S. No.	Assignment	Work stream	Duration	Deliverable	Status
1	[Assignment name]	[WS-_]	[]	[]	[]
2	[Assignment name]	[WS-_]	[]	[]	[]
3	[Assignment name]	[WS-_]	[]	[]	[]
4	[Assignment name]	[WS-_]	[]	[]	[]
5	[Assignment name]	[WS-_]	[]	[]	[]
6	[Assignment name]	[WS-_]	[]	[]	[]
7	[Assignment name]	[WS-_]	[]	[]	[]
8	[Assignment name]	[WS-_]	[]	[]	[]

TO COMPLETE: Fill this table with every assignment you worked on, including the Chapter 5 project. A completed assignment record is often the first thing a faculty evaluator reads.

6.3 Packaging and Collateral Beyond the Documented Project

TO COMPLETE: Describe other packaging work — for each item, state the product, whether the dieline was new, adapted or supplied, the construction and substrate chosen and why, how many iterations it went through, and the outcome. Two or three short paragraphs is sufficient; depth belongs in Chapter 5.

6.4 Prototyping and Product Testing

Prototypes outside the documented project followed the same escalation logic, with a structured observation protocol during testing. Testing conducted was internal, exploratory and qualitative; it was not a controlled user study and is not presented as one. Each session recorded the prototype version, the task attempted, the observed outcome, the failure mode and its location, and the design change proposed in response. Recording the location of a failure matters because a repeated failure at the same feature indicates a design cause rather than a handling cause.

TO COMPLETE: Describe the prototypes you built or assisted with — what each was for, what method was used, what question it answered, and what it revealed. If you operated a 3D printer, state material, layer height and any print failures; that level of detail distinguishes a genuine account. Do not describe testing with children unless it took place, and if it did, state how consent and supervision were handled.

6.5 Marketing Creative Support

Creatives were built as component-based systems rather than flattened compositions, so a concept could be adapted across aspect ratios and placements without rebuilding. Digital creatives are viewed at small size on a handheld screen, are frequently cropped automatically by the host platform, and are constrained by marketplace rules on image background, text-in-image proportion and permitted claims. Product photography and renders were prepared once and reused across placements, with layout components repositioned per format.

TO COMPLETE: Describe the creatives you produced — marketplace listing images, social posts, catalogue pages, trade sheets, in-store collateral — and any platform-specific rules you had to design within.

6.6 Work Streams Not Undertaken

TO COMPLETE: If any work stream did not form part of your actual internship, state so plainly here rather than padding the section above. An accurate account of a narrower internship is stronger than an inflated account of a broader one.

CHAPTER 7: OBSERVATIONS AND LEARNING OUTCOMES

7.1 Status of Claims

This chapter presents three categories of content, each labelled: qualitative observations from practice, feedback received from the organisation, and a proposed evaluation framework specifying what would be measured were the design function assessed quantitatively. No figure is presented as a measured result.

7.2 Observations from Practice

1. Physical verification detects a class of error that screen review does not. Proportion errors, fold-crossing artwork and legibility failures at true size were repeatedly caught on paper mock-ups after passing screen review.
2. The cost of a change is a function of the stage at which it is made — hours at concept, a production cycle after tooling commitment.
3. Constraints consulted early become parameters; constraints discovered late become revisions.
4. Mandatory content must be placed first, or it occupies residual space and becomes the weakest element on the pack.
5. An instruction guide cannot be validated by its author; only observation of a naive assembler reveals ambiguous steps.
6. In a growing brand a reusable template repays itself across the range while a one-off does not.

7.3 Challenges Encountered

Table 13: Challenges encountered and responses

Challenge	Response adopted
Screen-to-print colour shift on brand-critical elements	Working in the destination colour space from the outset; brand colours specified as spot references with documented equivalents
Artwork distorting or breaking across folds	Panel-by-panel development on the dieline; mandatory physical fold before design review
Manufacturing constraints surfacing after artwork completion	Consulting manufacturing at structural definition rather than at review
Instruction steps clear to the author but ambiguous to a first-time assembler	Naive-assembler validation; one action per step; fixed viewpoint
Statutory content competing with creative layout for space	Mandatory content placed on the dieline before creative layout
Risk of an unapproved file reaching a vendor	Status tokens in file names; only PRODUCTION-status files released
Vendor lead times compressing available iteration time	Front-loading low-fidelity iteration; escalating fidelity only when required

TO COMPLETE: Add challenges specific to your experience — software you had to learn, a failed print run, a prototype that would not assemble, a brief that changed mid-project. Specific difficulties honestly reported are the most persuasive evidence that the internship was real.

7.4 Feedback Received

TO COMPLETE: Summarise feedback in both directions — what was consistently identified as a strength and what was identified as needing improvement. A report that records critical feedback honestly reads as more credible than one recording praise only. Attach any written evaluation as an annexure.

7.5 Proposed Evaluation Framework

Quantitative measurement was not part of the internship. The framework below specifies what would be measured were the design function evaluated systematically. Every metric is marked as not measured, and none of these are reported as observed values.

Table 14: Proposed evaluation metrics (not measured during this internship)

Metric	Method of measurement	Status
Revision count to approval	Count from the version-control archive	Proposed
Pre-press defect rate per file released	Vendor pre-flight reports	Proposed
Assembly success rate without assistance	Structured observation with a defined sample	Proposed
Cube utilisation and dimensional weight	Dimensional calculation	Proposed
Template reuse rate across new projects	Asset library audit	Proposed

Revision count to approval and assembly success rate would be the most informative and the least expensive to institute, since both derive from records that already exist or from a low-cost observational protocol.

7.6 Skills Developed

Table 15: Competencies developed during the internship

Category	Competency
Technical	Dieline construction and adaptation; line-type conventions; bleed, safety zone and glue flap design
Technical	Print production literacy — colour space management, separations, substrate and finish specification, pre-press verification
Technical	Physical model making and additive prototyping
Technical	Design-for-manufacturing analysis against tooling, material and assembly constraints
Domain	Labelling and conformity obligations for toys, and the principal international standards for export
Professional	Working to production deadlines set by vendor lead times; operating within a confidentiality undertaking
Professional	Version control, documentation discipline and written approval records

7.7 Relationship to the Academic Programme

The internship was undertaken in a design and manufacturing context while the academic programme is in computer science and engineering. The connections are substantive rather than rhetorical.

- Constraint satisfaction. Packaging design is a constraint satisfaction problem with interacting hard and soft constraints — dimensional, material, cost, regulatory and aesthetic — and the reasoning parallels constrained optimisation.
- Release management. The naming, versioning and release discipline applied to design files is structurally identical to source control practice, with the same failure mode: an unapproved artefact reaching production.
- Usability and human–computer interaction. Assembly guide design is an information architecture problem, and naive-assembler validation is a usability test conducted on a physical artefact.
- Pipeline design. Establishing a workflow with defined gates, exit criteria and feedback loops is process design, and the same reasoning governs build pipelines and quality gates in software delivery.

TO COMPLETE (recommended): Name the specific subjects from your syllabus that connected to this work — for example Human–Computer Interaction, Software Engineering, Computer Graphics or Design Thinking. Evaluators often ask an interdisciplinary intern to justify the placement, and naming actual courses answers that directly.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This report has documented an industrial internship undertaken as Graphics and Prototyping Intern at Intelligientia Labs Pvt. Ltd., working on the InventoJoy toy brand from 26 March 2026. The engagement spanned packaging and dieline design, prototyping and product testing, instructional design, design-for-manufacturing collaboration and marketing creative support, and is documented through one project traced end to end together with a record of the wider work undertaken.

The central professional learning is that in a manufacturing context the quality of a design outcome is governed less by the merit of individual creative decisions than by the order in which constraints are admitted into the process. A workflow that establishes structural, manufacturing and regulatory constraints before creative development, and that verifies decisions physically at low cost before escalating fidelity, produces fewer revisions and fewer late failures than one treating those constraints as review criteria applied to a finished proposal. This differs materially from the way design problems are framed in coursework, where the constraint set is usually given rather than discovered and where a late change costs a revision rather than a production run.

8.2 Contribution to the Organisation

TO COMPLETE: State what your work contributed in terms you can substantiate — production-ready artwork sets delivered, a template or dieline reference established, an assembly guide validated before release. Avoid claiming commercial outcomes such as sales impact.

8.3 Future Scope

8.3.1 For the organisation

1. A formal dieline library with parametric templates for standard constructions, reducing structural definition time on new projects.
2. A documented brand and packaging design system covering colour standards, typographic scale, iconography and panel grids.
3. Institution of the metrics proposed in Table 14, beginning with revision count to approval and assembly success rate.
4. A pictogram library for assembly guides, giving the range a consistent instructional language.
5. Sustainable packaging evaluation — mono-material construction, reduced lamination where scuff performance permits, reduced void volume.
6. Computational tooling for repetitive tasks: automated generation of standard dielines from product dimensions and scripted pre-press verification.
7. Augmented-reality or video assembly guidance as a supplement to printed instructions for higher-complexity products.

8.3.2 For the student

1. Deepening parametric and computational design capability, connecting the design domain to the computing programme.
2. Formal study of the applicable toy safety standards and conformity assessment procedures.
3. Structured usability testing methodology, applicable to both physical and digital instructional artefacts.

8.4 Final Remarks

An internship in a growing consumer brand provides exposure that is difficult to reproduce academically: a small team, a broad remit, real production consequences and immediate feedback. The value of the engagement lay less in any single deliverable than in the shift from treating design as a problem of proposal to treating it as a problem of delivery — a distinction that becomes legible only when a decision has a cost attached to it.

REFERENCES

The structure below is provided for completion. No citation is presented as verified. Each entry must be completed with the actual title, publisher, edition or version and access date of the source consulted, and any entry not actually consulted must be deleted before submission.

ACTION REQUIRED: Replace each bracketed entry with a source you have actually read, formatted in the citation style your department requires (commonly IEEE or APA). Delete the rest. A short accurate reference list is always better than a long invented one.

A. Standards and regulatory sources

- [Bureau of Indian Standards — Indian Standard on safety of toys; insert exact designation, part, edition year and title as verified from the BIS catalogue.]
- [Government of India, DPIIT — Toys (Quality Control) Order; insert exact title, year and gazette reference as verified.]
- [European and North American toy safety standards, if relevant to export work; insert exact designations and years as verified.]
- [Packaged commodity labelling rules applicable to retail packaging; insert exact title and year as verified.]

B. Technical references

- [Reference work on packaging structural design and carton construction — author, title, edition, publisher, year.]
- [Reference work on print production and pre-press for designers — author, title, edition, publisher, year.]
- [Reference work on design for manufacturing and assembly — author, title, edition, publisher, year.]

C. Software documentation and organisational sources

- [Documentation for each design and CAD application used — publisher, version, URL, date accessed.]
- Intelligentia Labs Pvt. Ltd. Appointment letter for the position of Graphics and Prototyping Intern, InventoJoy. Summarised in Appendix A.
- [InventoJoy brand guidelines and vendor specification sheets — internal documents; cite only if permitted by your reporting manager.]

APPENDIX A: INTERNSHIP ENGAGEMENT DETAILS

Reproduced from the appointment letter issued by Intelligentia Labs Pvt. Ltd.

Table 16: Appointment letter particulars

Item	Particulars
Issuing entity and brand	Intelligentia Labs Pvt. Ltd. — InventoJoy
Address	E-003, First Floor, EPIP Site-5, UPSIDA, Greater Noida – 201306
Position	Graphics and Prototyping Intern
Effective date	26 March 2026
Duration and probation	Minimum 3 months; 7-day probation
Working hours	10:00 – 19:30, or as per operational needs
Work location type	Work from office
Review	Periodic, by the reporting manager
Notice period	30 days
Confidentiality	NDA covering customer data, business strategies, source files, assets and operational processes
Non-compete	6 months post-departure; e-commerce and toy sector; within India
Progression	Pre-Placement Offer may be considered subject to performance
Signatory	Director, InventoJoy

NOTE: The appointment letter also specifies stipend and incentive terms. Departments differ on whether financial terms belong in a submitted report — many prefer they are omitted. Add a row only if your department requires it, and consider redacting the compensation clause if you attach a scan of the letter. Attach the signed appointment letter and the company completion certificate here if required.

APPENDIX B: WEEKLY LOGBOOK

Table 17: Weekly internship logbook (to be completed)

Week	Dates	Work undertaken	Skills applied or learned	Supervisor initials
1	[]	[]	[]	[]
2	[]	[]	[]	[]
3	[]	[]	[]	[]
4	[]	[]	[]	[]
5	[]	[]	[]	[]
6	[]	[]	[]	[]
7	[]	[]	[]	[]
8	[]	[]	[]	[]
9	[]	[]	[]	[]
10	[]	[]	[]	[]
11	[]	[]	[]	[]
12	[]	[]	[]	[]
13	[]	[]	[]	[]

APPENDIX C: PRE-PRESS AND DFM CHECKLISTS

These checklists were applied as gates rather than as review by inspection, because the errors they catch are precisely those that inspection misses.

Table 18: Pre-press verification checklist

#	Check	Verified
1	Document dimensions match the approved dieline exactly	[]
2	Bleed applied on every cut edge; no critical content inside a safety zone	[]
4	Dieline on a separate named layer set to suppress at output	[]
5	Cut, crease and perforation lines on their correct spot colours	[]
6	Colour mode correct; no unintended RGB elements remain	[]
8	Overprint and knockout settings verified in separation preview	[]
9	All placed images at adequate effective resolution at final size	[]
10	All links present and packaged; fonts embedded or outlined as required	[]
11	Minimum type size respected, particularly for statutory text	[]
12	Barcode correct, at valid magnification, with quiet zones intact	[]
13	All mandatory statutory content verified against the approved content list	[]
14	Age grading, warnings, conformity mark, manufacturer details and origin present	[]
15	Glue areas free of varnish, lamination and heavy ink coverage	[]
17	File named with correct version and PRODUCTION status token	[]
18	Written approval of this version recorded before release	[]

Table 19: Design for manufacturing checklist

Area	Check	Verified
Dimensional	Internal dimensions accommodate the product plus tolerance	[]
Dimensional	Board thickness accounted for at every fold and internal clearance	[]
Structural	Glue flap dimensions adequate for the specified board grade	[]
Structural	Physical mock-up folded and verified against the actual product	[]
Material	Finishes compatible with substrate and clear of glue areas	[]
Process	Dieline nests efficiently on the vendor standard press sheet	[]
Process	Carton can be erected and packed on the available line	[]
Product	Tolerance stack-up assessed; part count reviewed for consolidation	[]
Logistics	Primary pack fits the master carton without excessive void	[]
Compliance	Statutory content accommodated on the dieline before creative layout	[]

APPENDIX D: IMAGES REQUIRED AND PRE-SUBMISSION CHECKLIST

Table 20: Images required to complete this report

Figure	Image required	Section
Fig. 1	Annotated generic dieline showing cut, crease, bleed and safety zones	3.3

Figure	Image required	Section
Fig. 2	End-to-end design and prototyping workflow flowchart	4.1
Fig. 3	Primary carton — flat dieline with line types annotated	5.2
Fig. 4	Insert tray — dieline and folded assembly with parts loaded	5.3
Fig. 5	Applied artwork on the dieline and assembled carton views	5.4
Fig. 6	Prototype iteration sequence, first mock-up to final	5.6
Fig. 7	Assembly guide — parts inventory page and step sequence spread	5.7

Table 21: Pre-submission checklist

#	Item	Done
1	University, department, programme and academic session inserted on the cover page	[]
2	Internal and external guide names inserted on the cover and certificate pages	[]
3	Company completion certificate obtained and attached	[]
4	Every grey placeholder box completed or deleted	[]
5	All seven figures inserted, with written permission obtained for proprietary images	[]
6	Chapter 5 aligned to your actual project — product, dimensions, materials, outcomes	[]
7	Assignment record and weekly logbook completed and signed	[]
8	Reference list completed with actual sources in the required citation style	[]
9	Standard designations in Section 3.8 verified against a current source	[]
10	Table of contents, list of figures and list of tables updated in Word (Ctrl+A, then F9)	[]
11	Margins, spacing and page numbering checked against departmental formatting rules	[]
12	Report proofread; plagiarism check run if required	[]